ActInf Textbook Group ~ Cohort 8 ~ Session 7 (Chapter 4) ~ 4/3/2025

TextbookGroup/ParrPezzuloFriston2022/Cohort_8 | Cohort 8 ~ Session 7 (Chapter 4) ~ 4/3/2025 | 1:06:01

All right, welcome back to cohort 8. So
we are in the first discussion of
chapter
4. Where would anyone like to begin?
Could be any section of four or any any
question or where four fits in or
just anywhere anyone wants to be in
first.
Yes, Sophia.
Um, I I found this chapter quite heavy
in math and I'm I'm still working
through it. I'm just wondering like how
much of the math is required to know for
like modeling or for being able to go
through papers in the field of
inference. I I suppose we don't have to
like know everything by heart but like
which figures or equations do you think
would be very essential to go
through to be like the foundation for
understanding at
inference? No. What does anyone think on
that?
I'm sorry, I was looking through the
textbook.
Um, seems like a question for
I I I think it's a super interesting
question. One piece that comes to mind
is like how much math would be required
for doing
uh
neuroiming like SPM modeling or just any
other kind of statistical using a linear
regression in Google Sheets or Excel or
R or

Python or using uh uh some sort of
statistics. It's like there's a lot of
math that that stems from and describes
and kind of gives rise to those
statistics equations and and same could
be said um for going into the like you
know what gives rise to the linear
algebra or the probabilistic systems
part. So it's in it it's like this this
is a really a critical and kind of that
that is the question like we get to
chapter 4 and it's like okay this is
where the the two roads came in from
both sides and now this is where it
actually is going to be all right this
is the key piece but how much do we need
to know in order to to do different
things and just how much of this is
um perhaps from like uh just just
background. Even though it's hard to say
what what's the uh background and what's
the key piece because there clearly this
chapter was written to to try to at
least give a
uh some balanced but minimum
uh blend of of true
background
topics like fundamental properties of
probability
distributions logarithms and derivatives
[Music]
um probabilistic
models probabilistic models that have
reference with about observations and
actions. So we can continue going on,
but I think
like th those kinds of statistical
models could be applied and

useful w with or without understanding
or being able
to to know all of
this.
Um, however, it's it's good and
and kind needs to be at least
understandable how the tools and the
implementations relate to what
their formal structures
are. What what what does anyone else
think about it
though? because definitely that wave
kind of hits hard in chapter 4 with with
just just being kind of half
equation-y. I I don't know. My my my
personal feeling is that the the
equations and the sort of computer
modeling seem to be kind of disconnected
from each other in practice. I just get
that feeling.
Do you feel that way for like using
linear regression models like just using
statistics and papers the way that P
values are used? I'm I'm just curious.
That's kind of just where I calibrate
the question to. Uh no, not not in the
same way because um all these equations
are supposed to like represent things in
thermodynamics and you know I don't see
any uh thermodynamics in like the
computer models.
Yeah, I think you know this is something
that that's um a uh an interesting
ambiguity or sort of
juxtaposition or like a a namespace
crowding with like energy based
models. So there's like energy as as the
potential and the kinetic energy of of

some object and mass for example.
um and and this sort of sense of using
an energy function to to fit some
statistical model. And then there's like
if, whether, how that actually connects
to
thermodynamics like candles burning and
um the energy density and like
temperature
dynamics, but it's possible to make
these energy based
models that that but it's not talking
about like energy in the gasoline
burning, but that's kind of
like they they
they are in different domains, right?
Yeah. Well, so so I understand that the
word energy and AI can mean all kinds of
things. So even in math, like for
example, there's something called the
deer energy, which you know, but um I I
my understanding was that this was
supposed to tie into thermodynamics and
equations were supposed to be
interpretable as thermodynamics.
Yeah, I there's there's um there's a few
kind of
separable angles on that. So if you
follow the basian mechanics
um like Dalton Saka Devel and um that
line of work then it's
like generalization of
thermodynamics just in the same uh set
as the Newtonian the thermoinformational
and the quantum mechanics basian
mechanics being like um describing those
belief distributional
updatings and other so that's one sense

but that doesn't mean that it has to
connect to uh thermo in terms of
describing the energetics of systems it
still could just be an you could just be
talking about some Gaussian distribution
doing some
rotation it doesn't have to like tie to
the actual survival of some embodied
object
But then there's that that that second
angle is like is this relating to the
actual thermodynamics of systems and and
how they maintain themselves above and
beyond what um just not
dissipating which is like the kind of
autooesis.
That's just sort of how how even
qualitatively the biological systems are
framed. So there's but but that's a
little bit of a separate
um question almost or or area of of
applying it than just trying to
generalize the theoretical physics of
information systems.
Daniel, if I could jump in real quick.
Uh I really appreciate the way you bring
so many disperate fields together. Uh
and I also appreciated Sylvia's
question.
Um there's so many levels to talk about
this, but I think it's really important
always to circle back to the principle
because that's different than the math.
The the principle of free energy.
um at being a framework. This is where
I've come to
um un understanding uh that it's really
important to understand the

framework. A it is a framework that uh
is applicable in so many different ways.
uh and I had to go back and look up
Elmholtzian free energy and uh you know
the the the chemical free energy all in
terms of thermodyam thermodynamics they
uh can be contextualized under different
conditions but really the principle
provides a
framework to apply so many different um
fields which I think is the brilliance
of using this framework. It
allows using a structured set of
equations that really uh you could go
into with a lot of depth, but you have
to understand the language, the what the
symbols mean. I think this was sort of
getting at Sylvia's question. uh you
have to understand what the terms are,
what how to construct the equations in
order to u be able to construct
particular applications. But there are
so many tools to use
um uh in in terms of what John was
talking about you know the computer
modeling is more about classification
uh whereas the um mathematical equations
are just giving the rigor uh uh to to
the cont to the
context. Um I uh I I I want to hear you
talk about this more Daniel but
um uh bringing in those who want to
apply the principle to these different
applications whether it's on the
molecular level or the social level uh
that the the framework can be used. I I
I really think that's the value of the
free energy principle is it it's just a

a a common language that can be used in
all these different applications and if
u even if a person doesn't know what the
symbols of the math means you know they
in consultation with others they can
find that out but just the understanding
that it is a consistent universally
applicable framework of how systems
evolve in this free energy principle, no
matter how complex it
is. I I don't know if that's helpful,
but this is just my approach to
contextualizing all these different
layers, these different
applications. Uh it's centered around
this principle.
Um so on Helmholtz free energy so it it
looks like that the sort of fristen free
energy is um you know supposed to be the
same as in addition to other things the
helmholds free energy because because of
that uh energy and entropy term where
they talk about it being proportional to
to energy the proportionality of course
being temperature but
I so the the weird thing is that that
energy term comes from the Boltzman
distribution. And so I'm wondering if
there's a more general form of the
equation that can deal with other
distributions or and in particular one
thing I worry about is uh the the
Boltzman distribution is one one right
but if you have some distribution that's
not one one say a Gaussian uh then it
can be then trying to compute the energy
from the probability isn't going to work
because you don't have a one one

function. I I'm just questions like
that.
Yeah, lot lot to say just to these last
points. This makes me think of this
uh one paper which I'll put in there
like what makes
something what makes an equation an
energy
function? I I'm I'm I'm actually just at
some there's probably lots of levels to
answer this. What makes something an
function? What makes it a free energy
function? And then
like what what are all these different
flavors with adjectives or with last
names like oh this is this
approximation.
How many layers deep into qualifying
like what kind of it it
is matters for the
FEP and h how many how many flavors of
that matter for
applying like that does it are there in
practice across different settings
um or
or is it going to is it going to get
super tied down in the weeds with
needing to specify all
these really different settings for for
um and like slightly different
uh unjustifiable assumptions will lead
to very different approaches with
different
um different like computer
approximations or or will like uh
just multiple full pretty
uh different
uh approaches give give broadly similar

results. So for example in the SPM uh
textbook the there are chapters that
have elements with classical parametric
statistics basian statistics and
nonparametric statistics. So like the
bootstrap and shuffling and relabeling
techniques like procedural techniques to
randomization and these and the
classical and the basian form of
statistics and I and and probably things
like conformal or other statistics too.
It's like if um there really is a warmer
room and a cooler room and the
thermometer has nonzero
information, you pretty much should be
able to detect it.
somehow. So that's the that's the kind
of
empiricist part and I think that and and
this is a perfect chapter 4 question
because it's like okay low road was just
about making the
computational building and assembling
the joint distributions for basian
models.
High road is more theory and axiom
driven. So then it's like okay so how
um how
general or um or or relevant is is this
meeting
point? Is this like every single model
like even just a linear regression? You
could interpret that as as doing
some process describable like this. Like
does this
just redescribe and generalize
statistical methods or or or
what what what is really being said or

claimed beyond what
Well, you you asked the the precise
critical question. What are
the important
constraints in constructing your model,
one's model? What are the important
variables or or or factors to to
consider like are is the
noise randomly distributed and
independent or not? you know that
determines what kind of statistics
you're going to
use. You know, is it is it a constant
pressure system for Gibsonian free
energy or is it a constant temperature
system for uh what is it? Hamiltonian or
hem helmholtz you know it's uh could you
consider the activity of neurons in the
brain as a
uh more or less uh constant energy but
redistributed
um it it it really is a creative process
in figuring out how to construct the
problem and I I think that's why
uh it's helpful to understand here's the
framework. Here are the tools to use.
Now, how can we rearrange the problem uh
to make the appropriate assumptions
uh to use the formula in the proper way
and that's going to take a lot of I
think collaboration or no individual
brilliance but
uh collaboration would be more helpful
to me.
Yeah, those are great comments and it
reminds me a lot of complexity science
and system science like also that is
about a framework and approaches about

patterns and systems of different
scales.
um and also has long featured
agent-based modeling like using net logo
um and other agent-based modeling
approaches. So the that that is a big
part of it is
um and
then where where do
the so just beyond that qualitative
structural
um possibility of complexity science?
What do we really get with chapter
4? But it helps connect all the way to a
statistical like looking at um
neuroiming and SPM. And so thinking
about first in uh prefab working with
just this one defined data input type
like fMRI data um or or uh EEG data.
And so kind of constraining on what the
the input data
were and then seeking to make cognitive
inferences about perception and
action. So
that and then
um it it it kind of built many of the
empirical precursor threads like fitting
latent spa fitting in uh fitting
empirical data noisy empirical data
uh finitely. So that's kind of the
connection point with the with the
empirical nature of of data in in like
research
um with projecting onto latent
spaces whether just generalized
dynamical systems latent
spaces or or information spaces like
probability distribution

spaces which uh is a space where there
can be more
axiomatic constraints on like
thingness which in many situations can
be kind of like a just a tautology or or
an auxilary point or overkill related to
just statistics and
practice. But it gives a lot of touch
points with some of these statistical
information underpinnings which then tie
into at the very least like the
efficient ways to model and identify
systems
empirically and then I think there's
like a lot of going further than that to
this like against that second angle kind
of like to John's question isn't this
about the thermodynamics it's like but
isn't this really about information
processing not just doing um baze
optimal statistical modeling of
empirical
data like Chris Fields's course wasn't
about saying here's how you fit the
models that balance the right number of
parameters with the right not
overfitting the
data where like complexity minus
accuracy is just a
normal measure for not overfitting a
statistical model, but that doesn't need
High Road. Uh, or it doesn't need a
4. Here's the question. How would you
determine the actual free energy of say
I I have a cat here. How do you
determine the free energy of my cat?
It's like what's the what's the mean
squared error of your

cat? It's like a model specific. If
you're if you're looking for a
number, it it's just a feature
of we we can look at equation 2.5, but
it's not like it's defined outside of a
given
distribution as models,
right? Well, so one issue is that that
the distribution is the Boltzman
distribution, right? Because you you can
tell that from the energy equation. Um
but the Boltzman distribution assumes
that you have a closed system and that
the uh the energy is constant. But a cat
is an open system. His energy can change
when he runs around or eats. Um so like
I this is this is a thing my physicist
Fred brought up to me is how do you
determine how do you act how do you
actually use this on a complex
system? Uh John, real quick, that is
exactly the question that I ask so
often. How do you set the bounds of the
system? Uh clearly at ultimately you're
talking about the universe, but if
you're just talking about the cat, you
have to make certain assumptions for the
moment, you know, for as an
uh and I know you like complex systems
also but you can uh set the bounds of
the complex system
even as as temporary approximation but
but that those kinds of approximations
will only work if things are of the
right scale. So uh if the energy is
changing fast enough then you it's not
it's not going to be approximated by a
Boltzman distribution. Right.

Yes. And then you have the hierarchical
levels. I think this is also brought up
not in this chapter but uh you just
layer and layer and
layer your
models. Exactly. Great point. So there
there's probably a lot in what you said
there um John and like right the
identifying the right scale. So one sort
of uh practical note on this is like it
may just be constrained by the data
that's being obtained like it's just
fMRI data. So it's it's constrained
versus the more like open multiscale
inquiry and then just like like um
Stephen said like this is where
structural modeling and and nested
systems but also there there's that kind
of Bucky Fuller like starting with
universe idea and kind of be and then
discarding irrelevancies and like the
subtractive method to identifying
systems boundaries. Um and then just in
terms of model
identification that's where there you do
uh structural ju just like you could do
with a multi with clinical data like
they just make
nested the the model comparison amongst
um every single variable every joint and
then just ask does this model with more
parameters explain more? Does it have
more accuracy minus complexity g you
know given explaining the variance in
this data
set given the number of parameters and
then so but treating that as a model
selection model

inference task like model selection as
inference and then identifying so okay
so we have um we have data coming in on
a continuous spectrum
And then it's and then so compare how
many different ways to
um partition that. Should there just be
one Gaussian? Should it be a mixture of
Gaussian? Should it be mixture of nested
Gaussians? But in a given situation that
might be like
overexploring, but some of these are
pretty well typed
out at least in in theory. So being able
to do at least kind of broad
sweeping explore a few structurally
different hypotheses pretty
fast and and and it's a super rich
question. There's a lot to to explore in
theory, but in practice it might just
be e easy to
identify. But it's pretty fascinating
that there's also a lot to explore in
that question about like the general
systems theory and nested systems in
their in their general
um
interactions and and that may lead to
some interesting new possibilities in
real world
systems. But in terms of of like what
data is available be like we have
um 10 replicates per
person from three
conditions. It's like well there's
that's that's the hierarchical level of
the statistical
model. So that's just more just making

statistical models that recapitulate the
act the levels of identifiable structure
in the
data. I I'd like to point something else
out which is
the the the the
the combination
of in the framework the a generative
model and um
uh using picking an appropriate general
generative model within this framework.
And just to point out in the brain there
are different generative models being
used like in the visual system there's
this convolutional network and in the uh
you know reward system basal ganglia
cortex you've got this reinforcement
learning generative model that they're
different types of deep learning models
but they can still be but they're still
within if in in our world they're
within this free energy, you know,
basian type most probable uh selection
process. I I just I guess my question
can be phrased as um if something I if
something if if my cat say wasn't acting
to to minimize his um to minimize his
fist and free energy, how would we know?
like what what test could we do to find
out uh whether he is or not?
And again, it's it's not really a
question of in my view
exactly that
um what is
um what is happening but rather how can
we
uh formulate it? How can we describe it?
And really all free energy does is

provide an upper bound
uh you know it's it's like um it's it's
accurate
enough for the task. It's complex
enough for the task it in terms of how
we describe it. We can make predictions
about what's happening. Um uh we can
model this system
as again using the brain as a reference.
Um there's a a a system of actors that
are reacting to what's happening in the
environment. And the way that they're
formulated
is in in
response to what's happening in the
environment, there are created these
priors and uh it in
uh behavioral uh interaction with the
environment. There are these uh
predictions of what's going to happen.
uh we say predictions but it's really
just the way the system is set up in in
a developmental response to how it has
been interacting with the environment
all along. You know nothing springs out
of nothing. It it's it's a developmental
process and it's this this is how the uh
the the free energy
formulation is a great descriptor of
that process. It's a great framework for
thinking about how that process occurs.
Um, uh, I get that you make models and
then you use the models to make the
predictions, but if you if you want to
make a model of my cat VV, its free
energy, you're going to have to try to
figure out what his free energy is. And
that brings me back to my previous

question of how do you figure out the free energy of a cat? Well, what's your model of a cat, I think, is the Um, I would like to I I would like to be able to uh predict the cat in the future. It's what you said. You you have predict to predict just the body temperature or blood sugar or the location or cognitive let's say location because that that that one's complex enough that it it bring it brings up the problem. How like Yeah. How do I know where he's going to run to? So that's probably doable actually. Yeah. So you have a location in two dimensions in the space. There's some distribution over X and over Y or there's some probability kernel summing to one spread out over the the the zone that you're going to be modeling this. So then it's like there's some diffusion process or there's some other underlying stochastic process giving rise um to samples coming from some true underlying latent distribution. And then it's like if you're using GPS or a video or just spot sighting, you're getting some empirical observations of the location of that cat. But but what if the so but if you do that like what if the variance is too high to do anything useful with it? Yeah. Big picture the whole point of statistics as a framework is so we can differentiate between like where do we have a super tight Gaussian and where do we have one that's so washed out that it's just like a flat line like it has infinite variance or something like

that. So big picture that's the whole point. be able to do estimates of variance like hey we haven't sampled enough to reduce our uncertainty and that doesn't mean that with some other prior or some other framework or or or framing there isn't information on data set it just like those observations hitting that distribution resulted in a wash out which could be a false uncertainty so it's not that any model is going to be like etc no free lunch but that that is the doing of statistics is determining the information content and and the relevance of the data on updating your prior and understanding whether you are just getting your prior back um or whether you're actually picking up with variable starting positions in prior families similar tendencies in in data.
But for something like a cat in order to get a a variance that would be low enough to actually do something with you have to like model like the whole brain of the cat. And this would bring into bring in issues of how do you estimate the cat's free energy, right? No, you might just have a a a a spatial temporal kernel learnable through any number of means including just all kinds of methods. Deep learning, parametric statistics. There might be so many ways to reduce your uncertainty by many bits about where the cat's likely to be. You can make a transition matrix by room and maybe just there's a circadian cycle.
Now where that exists between providing

no information like it's its behavior is
encrypted like maximum or whether it's
super recurrent with a high information
learnable
content. That's just an empirical
question. Well there's no there's no
there's no ability in modeling any given
situation or whether there's a given
signal for you in any given data set
that that it it that it's just it
doesn't take down statistics as a
framework or bypass it. It just means
there are situations that are with high
identifiability for certain modelers or
not. Well, well, I think the idea that
his behavior is encrypted is sort of
accurate because animals are supposed to
be unpredictable to other animals. And
so for that reason, I think you you
would have a very high variance and you
wouldn't actually be able to model it
unless you were able to model all the
sub aspects at the lower scales. Uh and
then that sounds really really hard.
Well, John, uh, it needs to be
unpredictable enough. It it needs to be
also conversely predictable enough. Like
there may be vast stretches of time
where they really don't want to be
predictable, but then there are other
local stretches of time where they want
to be very predictable, like when the
food is going to be there. So, there are
going to be
uh uh again, you don't have to know
everything to make the model. uh they're
going to be black boxes. Well, well, so
in chaos theory, if you want to model

something linearly in time, you have to
have exponentially more information.
Again, this was Daniel's point. It
depends on how, you know, uh how fine
grained you need to be.
Yeah. Let's say I want to know what
square foot he's going to be in for the
rest of the day. Yeah. Yeah. Like what
what performance you
get with what input data and what
spatial grid and and what kind of
reliability you need in the statistical
model. the that those are the exact
questions that are important for
statistics um in practice like data
engineers
um behavioral researchers like doing
statistics and determining I'm probably
all other kinds of data analysis
questions but this this is the meta
science of the approach to to have an
open-endedness
structurally in these settings and and
uh just practically to explain data.
Well, and and in practice to use a a a
multitude of statistical and
non-statistical methods
and also uh sampling methods. Like if
you want to know the square foot
location of a cat, you're going to have
to at least sample, you know, uh half
square foot resolution of the entire
space, you know, the closed system.
Well, I I think you'd have to sample
more than just location. I think you'd
have to do all sorts of sampling about
about the cat's nervous system as well
to to get the variance anywhere near low

enough to use for anything.
So that that that's again an empirical
question. Are you going to have is your
data just is your model constrained to
just sampled location data in latent
location data inferred you know andor
are you opening up this structural
possibility of well what if we do latent
state inferences like what if we also
have on this other side information do
we want to include temp temperature in
the room or do we want to include the
season like what other variable should
we include how should we mix and match
and compose and and flow these
variables the all the hit it hitting it
with the all by all statistical
analysis which it it might be doable in
certain settings but it's it you're
right that it scales computationally
poorly whereas for interpretability and
efficient computing the sort of
um goal approach is a sparse model where
where the sparity corresponds to like
the sort of cutting nature at the
joints.
doing actual relevant identification of
ontologically real
systems. you know, caveat, you know,
what is real, what is useful, but just
aiming somewhere in that useful
direction and in a less and less
uh predisposed more and more
structurally flexible way do multiscale
identification knowing that analytical
systems identification a certain way
would be implementable.
just with certain just you know knowing

that there are computational tools to
implement the
models beyond just sort of well you know
let this variable represent this and let
this represent this and kind of just
purely going off into a simple
direction. I I'm I'm just wondering if
the free energy principle can be tested
on animals or if it's just not
computationally going to be possible to
do
that. Like does applying it to choice
behavior counts or are you talking about
like the the what the textbook calls the
the whole iguana?
It's like I don't know. I I I I guess I
guess I would be happy with seeing
something impressive that couldn't be do
done by another means of analysis.
Well, if I may, this is it's also
applied to social systems. I mean, you
know, it's there are several papers on
uh psychiatric
uh uh
applications and it seems to be pretty
successful. I think, correct me if I'm
wrong, Daniel, but I think baked into
this approach is the probability
of the
uh how important you know the
probability that unknown variables
are important or not.
uh um you may not know all the var you
may not be able to model all the
variables but well that's just sort of
the variance right
well yes but there are different kinds
of variances uh and and

I well yes different kinds of variances
you're right that's that's what we're
talking
about that are being um iteratively
uh adjusted uh the
you know,
updating the priors in order to get a
more accurate probability spread, you
know, uh,
complexity, the the variational
probability.
I just wanted to jump in as as a novice
here and you mentioned the priors and
you also mentioned other approaches. I'm
wondering if this is um I guess it would
be an application of of free energy as
opposed to another approach if
you're doing updates
um based on
uh new observations that generate
prediction errors. But I I'm
thinking looking at my cats
um how how much do you weigh just
priors? I mean, if if you were to just
ask me, hey, um, you know, as as a
relatively lay person in in this
discussion, like where do you think your
cats are going to be in in a given time?
I say, well, what time of day is it?
This is where they are. This is where
the sun is. And so, like, a lot of the
variables seem seem kind of predictable.
So, then you've got this really strong
prior and your generative model is
probably fairly predictive. And then I
guess the next thing is well then how do
you add to the complexity of some of the
unknown variables? Uh there's a loud

sound outside. Uh you know, things that are unexpected, right?
I'm I'm looking Yeah, that was that was a great point. And it's kind of like the figure
42 thinking graphically and what is that Lego toolkit of graphical assembly diagrammatic assembly of
uh observables and
unobservables. So it's like that's that's and then which and that kind of gets into abductive
processes. Um model
selection, structure
modeling, selection
amongst statistical
models but mo but especially once you can create unobservables or do latent state inferences you did I mean
arbitrary chains of latent
states that that can be transformations or interle leavingings or just different patterns like one pattern that that people used to model the brain like was there there's convolutional like
patterns structurally
um there are hierarchical predictive processing like
uh functions this is kind of like Mars level of
analysis like what are the the algorithmic and the cognitive and etc layers or what what is happening
And then how is the structural the real dendrites and and ga and everything. How is that structural
um mechanistic layer for for different embodied systems or a silicon chip or

whatever related to
the symbolic
diagrammatic
relationships of of um observables and
unobservables and different
transformations.
Yes. Uh, John, you you you brought up a
philosophical question. Actually, you
may think you're wanting to model your
cat, but there's no such thing as a cat.
there's a
population and uh you know a cat is a
member of the population but it doesn't
exist
independently. I mean, again, I'm
talking philosophically here, really.
Uh, you want to model everything about
your cat, but you're absolutely right.
To model everything about your cat,
you'd have to model the entire universe
or at least the entire, you know, world
that that cat in inhabits every place,
you know, including other cats
and, you know, everything interacts
with. But but but what I'm worried about
is like what if he what if he's not
lowering what if he's not acting to
lower his free energy right now? How
would we know? Um that that's my
fundamental question is uh maybe maybe
he's acting to lower his free energy and
maybe he is not. How do we know?
Well, isn't free energy just an upper
bound to
uh the difference
between the complexity and the accuracy
or the the divergence and the evidence.
You know, it's it's it's

not a precise number. It's just an upper
bound to what the uh difference between
the energy and the entropy is. Yeah.
Look, that's that's one great point. Let
me give another piece on this. Here's in
in um chapter
six complete class theorems state that
any statistical decision pre procedure
so whether it's um ruminative repetitive
maladaptive you know um just leading to
demise leading to peril externalities
failure may be framed as baze optimal
under the right set of prior
beliefs. So the whatever whether the cat
is is staying in one room or whether
it's between two rooms or whether it
what or whether it leads to its demise
that can be toddloically modeled as its
own as a distribution. So there's no
there's no sort of perceived or or
projected pathology on like human or a
cat system or any behavioral situation
you can dream of that that couldn't be
seen from a modeler's
perspective from the outside as a
statistical decision procedure.
Okay. So are you familiar with David
Deutsche's criterion of good
explanations?
No.
So, a a good explanation has to have
something called difficulty to vary. Uh,
which means you you you can't just
change all of the details. Uh- which
means that if you're you know, if you
had a negative experimental result, you
can't just change the details to get
another one. Um, like I if the if the

idea is you just go through the the
enormously large space of all
statistical models until you find one
that says the cat is minimizing its free
energy. Un unless there's some concrete
way to actually check whether that
statistical distribution applies to the
cat in some way that there there you're
not going to have difficulty to vary
with an explanation like that.
Yeah, it that's it's it's what if you
think it is it possible to talk about
the statistician reducing their mean
squared error for their predictive model
cat? Like do you extend that sort of
belief to all of statistical modeling
and just doing behavioral
science? Because that's that's a
separate avenue than asking what the cat
is really doing and taking this sort of
like
ontological like what is the what is the
territory really doing is a little bit
of of a of a
extension to the the the systems
identification
question. And so just it's it's just
what
what it's they're a little bit different
application
questions. It's just that it's just that
cats are sneaky and they're
temperamental. So they may they may
decide that they're not going to lower
their free energy because they want to
confound us. And so how would we know
that they're not?
So there's a ton of ways to think about

it, but just like from a complex system
like or system science, general systems
perspective, I just think there any
particular data point you throw, cats
are sneaky or this I think this will
happen or they're like this or I saw
this or I don't think that's possible.
It's like we're looking for something
that includes all of those
expressions, but just like but then this
this mushroom did this next to this
tree, so how could that be doing that?
It just those are all inquiry
directions, but the whole point is what
is the grammar of all of those
identifications and assertions that
you're just
saying in a
row. there's a bigger. So, it's nothing
nothing about cats specifically. It's
just what is the language that you're
looking for design space within with
those kinds of claims you're making
about your cat. Well, it's sort of where
does the probability distribution come
from and what what does basian inference
add that wasn't there before in trying
to figure out the probability
distribution?
Yeah, fair enough. I think you're asking
really good questions and I you try it
and see if it works.
Yeah, that's a great way to put it. And
also and I also think about like are we
talking about modeling a situation
relative to just vibing and not modeling
a
situation or what what or are we talking

about the difference between basian
statistics
and parametric statistics and
non-parametric statistics and deep
learning or is it methods comparison
within we're already in a probabilistic
modeling technical
um sort of
setting or is it a philosophical
qualitative discussion about the nature
of things that isn't really looking to
be just deflated or resolved to like
yeah what is the free energy of
marathoners at this mile it's like well
are we talking about the temperature
okay well here's the data set here's the
value it was 2700
okay let me let me put it this way so so
my question can be phrased like this if
if I'm trying to model something by
introducing some kind of probability
distribution and using that to model it.
What do I get out of using basian
mechanics that makes that better than it
would have been without? So, you know,
anyone could just start from first
principles and try to model things with
a probability distribution. How does
that modeling process change if if
you're using the free energy principle?
Yeah, it's a great question and I I look
towards more and
more really cool documented more
explainer paths through that and so
beyond just looking at the literature
and the citations for what are described
in this textbook.
Um one I'll just give two two two short

ways. One way is like as a perspective
and a a stance. This is like how
information is like using this as an as
as part of our toolkit of information
theory and understanding statistics.
It's like this is just how statistics
inference is. If your joint distribution
that you're looking at is just like just
it it it's a lens on what statistics
have already occurred. you just look at
the statistics in a a clinical study or
behavioral study. So in that
lens it's just a way we can think about
even those statistical analyses. But
then as we start thinking more into
um basian graphical
models, it it gives composable
modeling
tools and then it just
it you you can do statistics modeling
and have these diagrammatic tools. And
there's other situations where it's it's
convenient and more than adequate to do
um like to think more about a data
science flowchart like here we're just
going to push it through a neural
network and get the output. So that's
thinking like at a systems uh
engineering level and just letting a ton
parameters with a ton of uh models with
a ton of parameters just learn
relationships and
data. Here's an
explicit
probabilistic distributional
approach that has some other
features and and those features are it
plays well with other techniques, other

models. it it it
allows interaction
um translation between different
approaches.
Uh you can start with a a very coarse
grain approximation and update it um to
make
predictions in a way that can be
interactive with other approaches.
is my view of this is the advantage of
using this approach. It it gives you
more tools.
Yeah, I think that compatibility of free
energy with other, you know, I guess you
could say simpler models is is appealing
and
and also it seems that if you reduced it
to its simplest application, it also
holds up. So in other words, there's
there isn't something that's too simple
um to be addressed with with you know
active
inference. That's a great point that
with with simple dynamical systems we
can get into the trivial and the inert
like the function y equals 5.
It's like, well, that's overkill to have
a function to have just this line going
for for infinite time at five when we
could just say the number five. It's
like, yeah, but then it can it's kind of
like a it has all these other features.
So there's there's there are statistical
models that would be um useless or
overkill or
underkill. That's the whole model
selection question in theory and
practice. But in terms of the um latent

space
structuring, it can be
um a coin flip world or it can be all
this
structured latent spaces and and and
structural models with different state
spaces, different time concepts.
Can I suggest that that's actually a
criticism of active inference in the
free energy principle that it is so
available uh to u interpretive
description that it can describe
anything depending on how you set up the
equations and the models. I mean its
strength can also be a criticism of it
in a way uh because you you really have
to um trust I guess how the models are
set up and again always proof is in the
pudding so to speak that the the you
does it work
um I I have heard people say that it's
Um, I don't understand it. It's it's
it's like it you can arrange the
components of
the equations in a way that it's sort of
like h having
uh any model with enough parameters can
describe any anything. Um but the
advantage is it's predictive
uh and and uh and flexible enough
to bring together different models into
one system.
Yeah, this is one funny math like quote
to to the that point. Um, with four
parameters I can fit an elephant and
with five I can make him wiggle his
trunk. So it's like
um w with

modeling it's just really what is the
what is the role what is the person's
expectation of the role of any of this
of the low road or the high road? How
much work do people expect or what kind
of foundational grounding do people
expect in
statistics? What do people think is
under the hood of a t test or a p value
uh or a baze
factor? Do people know about the the
zcore and the p p value factor and
they're looking for other criterion for
signal
processing and empirical systems
identification or are we pulling back a
step like what is the role or the
function of any quantitative
measure in comparison with just choosing
not to do
Yeah. So if I have a a patient in the
clinic who can't walk straight uh and I
build a model that says that there's a
part of the brain that predicts
uh what forces are
necessary on certain muscles to get to
walk straight. And there's another part
of the brain that's interpreting the
sensory input that's uh letting the
person know if they're walking straight
or not. And I set up a model that would
predict there would be an error in one
component of this system. There would be
a a a damage to one component of the
system. I'm going to now go look in this
person and see if there is a part of
their brain or nervous system that's not
functioning properly or there's damage

to it. And voila, there is damage to this part of the brain. It that that's a use a useful use of that predictive modeling.

Yeah. Well, a lot of interesting uh comments and and aspects today. So, thanks for joining. You could uh come next week at the same time discussing chapter 9 i.e. on uh empirical data fitting. So that kind of relates to what we discussed a little bit today or see you in two weeks. So thank you. Bye.